Particle Physics 2 Standard Model: Lecture 5 : New Revolutions in Particle Physics: Standard Model

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AI-assisted companion edition prepared with Video2Book

Chapter 1

Gauge Fields, Interaction Scales, and Charged Weak Currents

Quantum electrodynamics and quantum chromodynamics look very different in the laboratory, yet both are gauge theories. In fact, nearly every interaction in the Standard Model is organized this way. The word *gauge* has a deeper meaning involving local symmetry, but we can postpone that machinery. For the moment we need a minimal physical description: a gauge theory contains Maxwell-like fields, conserved sources, symmetry transformations, gauge bosons, and rules for emitting and absorbing those bosons.

1.1 Maxwell Theory as the Prototype

1.1.1 Fields and potentials

The electromagnetic field has six components: three components of electric field and three components of magnetic field. The same information can be encoded, with some redundancy, in a four-vector potential

$$A_\mu = (A_0, \boldsymbol{A}), \tag{1.1}$$

where A_0 is the electrostatic potential and $\boldsymbol{A}$ is the spatial vector potential. In the electrostatic limit the electric field is obtained from the spatial gradient,

$$\boldsymbol{E} = -\boldsymbol{\nabla} A_0, \qquad \boldsymbol{F} = q\boldsymbol{E}, \tag{1.2}$$

while the magnetic field is

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{1.3}$$

The sign in the first equation depends on the standard definition of the potential; the board discussion is concerned mainly with the fact that fields are derivatives of potentials. [1]

Thus electrodynamics admits two equivalent descriptions: one in terms of $\boldsymbol{E}$ and $\boldsymbol{B}$, and one in terms of A_μ. The latter contains only four displayed components because different potentials can describe the same physical fields. That redundancy is the first hint behind the word gauge, though we will not use it formally yet.

[1] Editorial clarification: For time-dependent fields the complete relation is $\boldsymbol{E} = -\boldsymbol{\nabla} A_0 - \partial_t \boldsymbol{A}$. The lecture deliberately works at the simpler electrostatic level here.

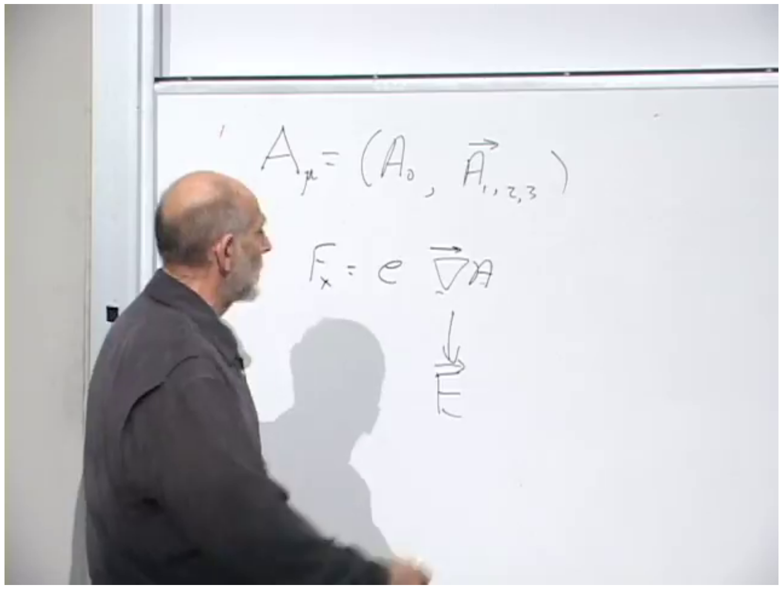

Figure 1.1: The electromagnetic four-potential and its derivative relation to the field.
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1.1.2 Waves and polarization

Maxwell's equations support waves. Take a plane wave moving along a unit vector $\hat{\boldsymbol{k}}$. At every point the electric and magnetic fields are transverse:

$$\boldsymbol{E} \cdot \hat{\boldsymbol{k}} = 0, \qquad \boldsymbol{B} \cdot \hat{\boldsymbol{k}} = 0, \qquad \boldsymbol{E} \cdot \boldsymbol{B} = 0. \tag{1.4}$$

In units with $c = 1$, their magnitudes are equal,

$$|\boldsymbol{E}| = |\boldsymbol{B}|. \tag{1.5}$$

For a linearly polarized wave, the direction of $\boldsymbol{E}$ specifies the polarization. The entire pattern propagates at the speed of light.

Quantization changes the language but not the polarization structure. A quantum of the electromagnetic wave is a photon. Drawing a photon as a point is inevitably misleading, but it is useful to imagine that the point carries a small transverse pointer recording its polarization. A quarter-wave plate or another optical element can rotate that polarization. A static Coulomb field may likewise be pictured, only roughly, as arising from repeated emission and absorption of photons.

1.1.3 Sources, flux, and conservation

Electromagnetic waves can propagate through empty space, but the field also has sources: electric charges and currents. A positive point charge produces an outward radial electric field proportional to $1/r^2$. The number of field lines piercing any sphere around the charge is independent of the sphere's radius. Mathematically,

$$\oint_{S^2} \boldsymbol{E} \cdot d\boldsymbol{S} = \frac{Q_{\text{enc}}}{\epsilon_0}, \tag{1.6}$$

in SI units.

This is more than a convenient way to calculate the field. Electric field lines may end only on charge. If one isolated charge simply disappeared, its field would either have to disappear everywhere at once, violating causality, or a front of missing field would have to propagate outward with field lines ending on empty space. Neither is allowed. The geometry of flux therefore makes charge conservation natural: charge can move, and opposite charges can annihilate, but net charge cannot vanish locally without a compensating current.

The corresponding local statement is the continuity equation

$$\partial_t \rho + \boldsymbol{\nabla} \cdot \boldsymbol{J} = 0. \tag{1.7}$$

[2]

1.1.4 A working definition of gauge theory

We can now state the operational pattern. A gauge theory has:

- electric-like and magnetic-like fields;
- waves that, when interactions are weak enough, resemble light waves;
- sources analogous to electric charge;
- a Gauss law tying flux to those sources;
- a conservation law and an associated symmetry;
- quanta of the field, called gauge bosons; and
- a coupling constant controlling emission and absorption.

The source need not be electric charge. As a deliberately unphysical example, imagine gauging baryon number. Protons and neutrons both have baryon number +1, so they would repel through a new Coulomb-like force; a proton and an antiproton would attract. Such a theory is mathematically imaginable, but nature supplies no observed long-range gauge field coupled to baryon number. Conservation by itself does not guarantee a gauge force.

Electric charge does come with a symmetry. For a charged matter field,

$$\psi(x) \longmapsto e^{i\theta} \psi(x). \tag{1.8}$$

The phase symmetry and charge conservation are two sides of the same structure. Later we can make θ depend on spacetime and see why the potential A_μ is required. Here the global transformation is enough to identify the conserved charge.

1.2 QCD as the Non-Abelian Example

1.2.1 Triplets, anti-triplets, and the gluon matrix

In quantum chromodynamics a quark field carries a color index $i \in \{r, g, b\}$ and transforms in the fundamental representation of $SU(3)$:

$$q_i' = U_{ij} q_j, \qquad U^\dagger U = \mathbf{1}, \qquad \det U = 1. \tag{1.9}$$

[2] Editorial clarification: Equation (1.7) is the standard local mathematical form of the causal flux argument developed verbally in the lecture.

This is a rotation in a three-dimensional complex color space, not a rotation of ordinary position space. The three-component representation is also called the defining representation.

Antiparticle fields are represented by the complex-conjugate variables. Taking the complex conjugate of Equation (1.9) gives

$$q_i'^* = U_{ij}^* q_j^*. \tag{1.10}$$

Thus quarks transform as $\mathbf{3}$, whereas antiquarks transform as $\bar{\mathbf{3}}$.

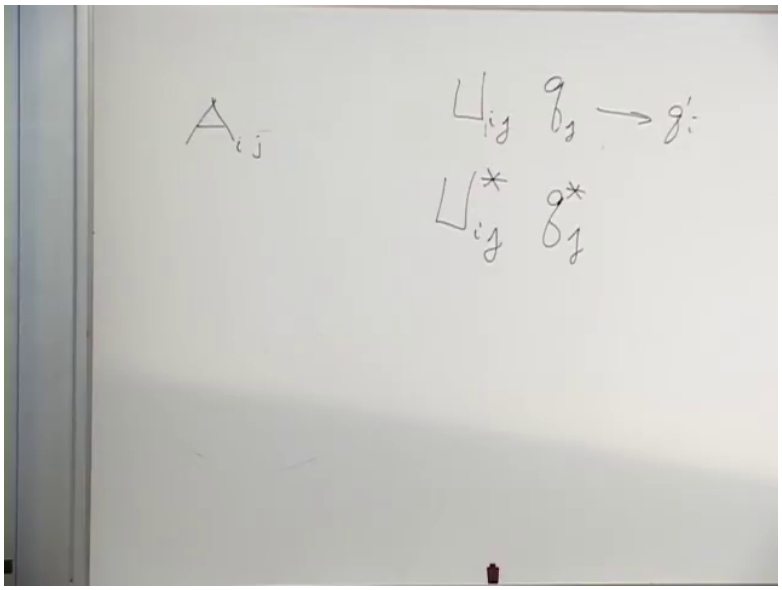

Figure 1.2: The fundamental transformation and its complex-conjugate anti-triplet transformation.
Lecture frame: 00:18:47.

Question from the audience. How many U matrices are there – seven, eight, or nine? [a]

Response. There are infinitely many group elements U. The finite number being counted is the number of independent generators, which is eight for $SU(3)$. A general 3×3 matrix has nine matrix positions, but special unitarity leaves eight continuous directions.

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[a] Lecture timestamp: 00:19:37.

The gluon field may be written as a traceless matrix A_{ij}. One index transforms like a quark and the other like an antiquark, so its color structure is that of

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}. \tag{1.11}$$

QCD retains the traceless octet,

$$A_{ii} = \operatorname{tr} A = 0, \tag{1.12}$$

and discards the singlet. That is why QCD has eight gluons rather than nine. More generally, $SU(N)$ has

$$N^2 - 1 \tag{1.13}$$

generators and the same number of gauge-boson fields.

The photon is neutral under electric $U(1)$: rotating the phase of an electron does not rotate the photon into a different photon. Gluons are different. A gluon carries a color and an anticolor, so it transforms in the adjoint representation. A red–antiblue combination, for example, is not a color singlet even though it contains one color and one anticolor.

1.2.2 Index flow and self-interaction

The interaction rule can be read by following indices. Schematically, a quark of color i can become a quark of color j while emitting a gluon carrying the corresponding color flow:

$$q_i \longrightarrow q_j + A_{ij}. \tag{1.14}$$

The precise placement of i and j depends on the arrow convention; the invariant content is that every index entering a vertex must continue through another line. No color index is lost.

This rule applies in any $SU(N)$ gauge theory. Fundamental matter has one index, antifundamental matter has one conjugate index, and a gauge boson has one of each. The physical names can change, but the index structure remains.

Because a gluon itself carries color, it can emit another gluon. An A_{ij} gluon can split through an intermediate index k, schematically,

$$A_{ij} \longrightarrow A_{ik} + A_{kj}. \tag{1.15}$$

This is not a complete Yang–Mills vertex, but it displays the color flow. Nothing analogous occurs for photons in ordinary electrodynamics. Gluons therefore exert forces on gluons, and non-Abelian field dynamics is much more complicated than Maxwell's linear theory.

One qualitative consequence is confinement. Color flux lines cannot end, and their self-interaction draws them into a tube joining a quark to an antiquark. The tube carries an approximately constant energy per unit length:

$$V(R) \simeq \sigma R. \tag{1.16}$$

Pulling the endpoints farther apart therefore stores more energy instead of liberating isolated color. This is the gooey piece of glue picture: the field itself prevents colored constituents from escaping as free particles.

1.3 Couplings and the Meaning of Strong or Weak

1.3.1 Charge as an emission amplitude

A gauge theory still needs a coupling constant. In QED the electron charge e is the coefficient associated with an electron–photon vertex. It is an amplitude, and probabilities are squares of amplitudes. Operationally, imagine stopping a fast electron abruptly in a cathode-ray tube. The probability for radiating a photon is proportional to e^2, with conventional factors absorbed into the fine-structure constant,

$$\alpha = \frac{e^2}{4\pi\hbar c} \simeq \frac{1}{137}. \tag{1.17}$$

In natural units $\hbar = c = 1$, the coupling is dimensionless. A one-percent scale for an elementary emission makes electromagnetism weak in the perturbative sense: most stopped electrons do not each release a spray of hard photons.

1.3.2 Natural units connect energy, time, and distance

Set

$$c = 1, \qquad \hbar = 1. \tag{1.18}$$

Then time and distance have the same units, while energy has inverse units:

$$E = \hbar\omega \longrightarrow E = \omega \sim \frac{1}{t}, \qquad x \sim t \sim \frac{1}{E}. \tag{1.19}$$

Large energies probe short distances and short times.

For orientation,

$$1\ \text{eV}^{-1} = \frac{\hbar c}{1\ \text{eV}} \simeq 1.97 \times 10^{-7}\ \text{m}. \tag{1.20}$$

An atomic length of order 10^{-10} m therefore corresponds roughly to an inverse kiloelectronvolt. These are order-of-magnitude conversions, not precision spectroscopy.

1.3.3 The hierarchy inside an atom

For a typical atom, take

$$R_{\text{atom}} \sim 10^{-10}\ \text{m}. \tag{1.21}$$

Light crosses this distance in

$$t^{\text{atom}}_{\text{transit}} \sim 10^{-18}\ \text{s}. \tag{1.22}$$

An electron, however, moves at only a small fraction of the speed of light in a light atom. In the Bohr picture $v/c \sim \alpha$, so a typical orbital time is longer:

$$t^{\text{atom}}_{\text{orbit}} \sim 10^{-16}\ \text{s}. \tag{1.23}$$

An excited atom takes longer still to radiate:

$$t^{\text{atom}}_{\text{decay}} \sim 10^{-9}\ \text{s} \tag{1.24}$$

as a broad lecture-scale estimate. There are two suppressions in the physical picture. The electromagnetic force produces only modest acceleration, and an accelerated electron has another small emission probability controlled by α. The small fine-structure constant therefore creates a hierarchy among transit, orbital, and radiative-decay times.

1.3.4 The absence of hierarchy in a hadron

A hadron is an atom-like bound system made of quarks and gluons, but its scale is much smaller:

$$R_{\text{hadron}} \sim 10^{-15}\ \text{m}, \qquad t^{\text{hadron}}_{\text{transit}} \sim 10^{-23}\ \text{s}. \tag{1.25}$$

Quarks cross the hadron on essentially the same time scale, and strongly decaying excited hadrons also live for roughly

$$t^{\text{hadron}}_{\text{orbit}} \sim t^{\text{hadron}}_{\text{decay}} \sim 10^{-23}\ \text{s}. \tag{1.26}$$

A struck hadron may vibrate and emit a pion almost as quickly as causality allows. There is no seven- or nine-order-of-magnitude separation like the one inside an atom.

Question from the audience. Does special relativity imply that a hadron cannot decay substantially faster than its light-crossing time? [a]

Response. That is a fair way to state the order-of-magnitude bound. Strong decays occur about as fast as the size of the system permits; there is no small coupling or other large suppression slowing them down.

[a]Lecture timestamp: 00:47:15.

The corresponding strong coupling is therefore much larger than α. At an illustrative hadronic scale the lecture quotes

$$\alpha_{\text{QCD}} \sim 0.2, \tag{1.27}$$

much closer to unity than $1/137$. This is the dynamical origin of the older name *strong interactions*. [3]

Question from the audience. Is the strong analogue simply called the strong-force constant? [a]

Response. It is conventionally called α_{QCD}, or more often α_s. The electromagnetic one is usually just α, though α_{QED} can be used when both appear in the same discussion.

[a]Lecture timestamp: 00:48:17.

Question from the audience. Would a paper distinguish a bare electromagnetic coupling from the measured, shielded value? [a]

Response. Yes. One speaks of a bare coupling and a renormalized coupling. Vacuum polarization makes the effective coupling depend on the scale at which it is measured.

[a]Lecture timestamp: 00:48:38.

1.4 The Puzzle of Weak Decay

Weak interactions were named from their very long decay times. They are usually slower than strong decays and can even be slower than atomic radiation. The oldest example is nuclear beta decay.

1.4.1 Neutron beta decay

A free neutron decays through

$$n \longrightarrow p + e^- + \bar{\nu}_e. \tag{1.28}$$

The lecture uses a lifetime of roughly twelve minutes, absurdly long compared with 10^{-23} s. Modern precision values do not matter for the argument.

Part of the slowness is kinematic. Roughly,

$$m_n \simeq 940 \text{ MeV}, \qquad m_p \simeq 939 \text{ MeV}, \qquad m_e \simeq 0.5 \text{ MeV}. \tag{1.29}$$

Only a tiny fraction of the neutron's rest energy is available as kinetic energy. If the neutron were lighter than the sum of the final rest masses, decay would be impossible. If it lay only infinitesimally

[3]Editorial clarification: The QCD coupling runs with energy, so α_s is not one fixed number. The quoted value is an illustrative scale-dependent value, as the lecture intends.

above threshold, the available phase space would be tiny and the decay would be slow. But phase space alone cannot explain a lifetime measured in minutes.

1.4.2 The charged pion removes the threshold excuse

Consider also

$$\pi^- \longrightarrow e^- + \bar{\nu}_e \tag{1.30}$$

and its charge-conjugate process. The pion has ample energy above the electron channel, yet its weak lifetime is still of nanosecond order rather than hadronic 10^{-23}-second order. A similar channel with a muon will soon become important. The puzzle is therefore intrinsic to the weak interaction, not merely an accident of the neutron's near-degenerate masses.

[4]

The term beta decay originates historically from calling emitted electrons beta rays. The pion and neutron examples involve the same charged-current interaction even though their kinematics differ sharply.

1.5 A Table of Quarks and Leptons

1.5.1 Color acts vertically

Arrange the six quark flavors in three paired columns,

$$(u, d), \qquad (c, s), \qquad (t, b), \tag{1.31}$$

and place a red, green, and blue copy of every quark in separate rows. Color $SU(3)$ acts vertically: it rotates red, green, and blue versions of an up quark into one another, and performs the same color rotation simultaneously on down, charm, strange, top, and bottom. It does not turn an up quark into a down quark.

1.5.2 Weak isospin acts horizontally

The weak symmetry acts in the other direction. It mixes each neighboring pair

$$Q_1 = \begin{pmatrix} u \\ d \end{pmatrix}, \qquad Q_2 = \begin{pmatrix} c \\ s \end{pmatrix}, \qquad Q_3 = \begin{pmatrix} t \\ b \end{pmatrix}. \tag{1.32}$$

The up-type quarks have charge $+2/3$, and the down-type quarks have charge $-1/3$, so the charge changes by one unit across each pair. [5]

Because the symmetry acts on doublets, the natural group is $SU(2)$. It acts on every color copy of each quark doublet without changing color.

[4] Editorial clarification: The charged-pion lifetime is about 26 ns, and the dominant channel is $\pi^- \to \mu^- \bar{\nu}_\mu$. The electron channel is helicity suppressed. The lecture first uses the electron channel because its particle bookkeeping is simplest, then returns to the dominant muon channel.

[5] Editorial clarification: The source momentarily says that a down quark has charge $-2/3$. The surrounding subtraction and every subsequent process require the standard value $-1/3$, used here.

There is also a fourth row of particles that do not carry color and therefore do not interact with gluons: the leptons. They form three more doublets,

$$L_e = \begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \qquad L_\mu = \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \qquad L_\tau = \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}. \tag{1.33}$$

The neutrinos are electrically neutral; the electron, muon, and tau have charge -1. Again the two members differ by one unit of electric charge. The electron, muon, and tau have the same gauge quantum numbers but different masses, just as the three generations of up-type or down-type quarks repeat the same charge pattern.

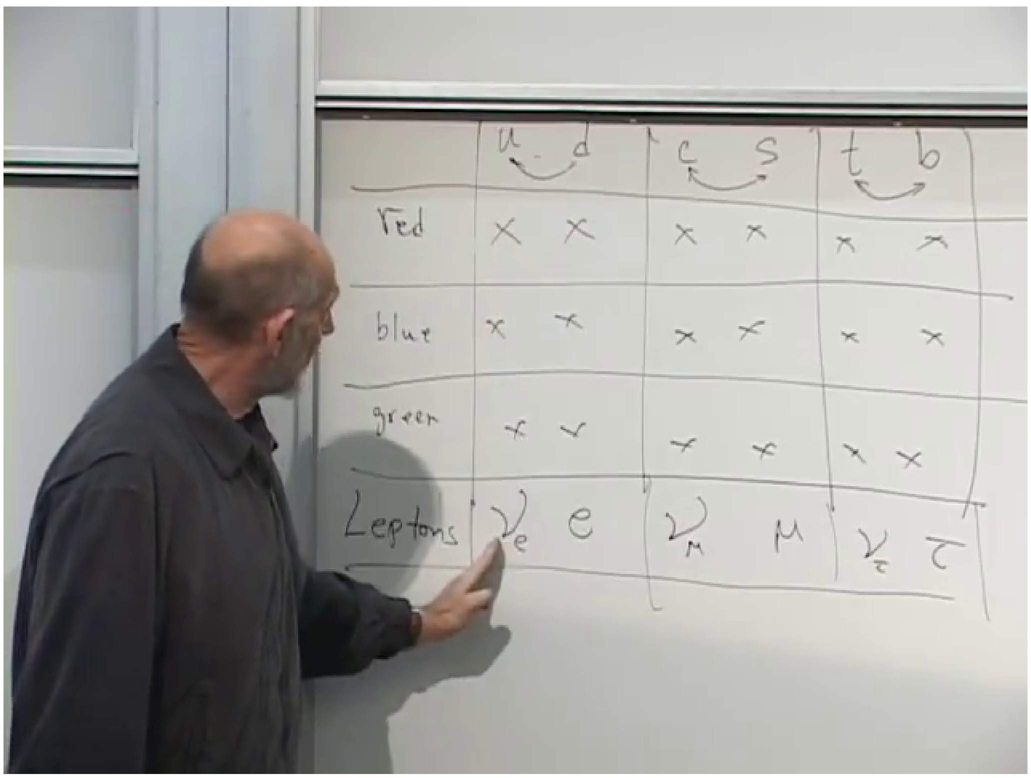

Figure 1.3: The board table: color acts among the red, blue, and green rows, whereas weak $SU(2)$ links the paired flavor columns and the lepton row.

Lecture frame: 01:03:35.

Calling leptons a fourth color is only an analogy at this stage. They fill a fourth row in the weak-doublet table, but ordinary QCD $SU(3)$ does not rotate quarks into leptons.

1.6 Constructing the Charged Weak Bosons

1.6.1 Quantum numbers of the charged bosons

Gauge bosons carry the transformation properties needed to move from one member of a doublet to the other. Begin with the lepton pair $e^-\bar{\nu}_e$. It has electric charge -1, so the corresponding charged gauge boson is called W^-:

$$W^- \sim e^-\bar{\nu}_e. \tag{1.34}$$

The same quantum numbers arise from a down quark and an anti-up quark:

$$W^- \sim d\bar{u}. \tag{1.35}$$

The symbol $\sim$ means has the same relevant gauge quantum numbers as. It does not mean that the W is literally a bound state of these particles. The antiparticle has the conjugate combinations,

$$W^+ \sim e^+\nu_e \sim \bar{d}u. \tag{1.36}$$

The charged weak vertices include

$$e^- \longrightarrow \nu_e + W^-, \qquad \mu^- \longrightarrow \nu_\mu + W^-, \tag{1.37}$$
$$d \longrightarrow u + W^-, \qquad s \longrightarrow c + W^-, \tag{1.38}$$

with analogous transitions for the third generation when kinematically allowed. The gauge boson performs dynamically the same horizontal mixing that the symmetry performs on the doublet.

Question from the audience. Charge conservation fixes the emitted boson to have charge -1, but why must the electron, down-quark, and strange-quark transitions all involve the same particle? [a]

Response. That is an empirical fact encoded by the common weak symmetry. Nature could consistently have assigned a different W-type boson to every family or to quarks and leptons separately. Instead, one $SU(2)$ acts simultaneously on all the doublets, and the same charged gauge bosons mediate all these transitions.

[a]Lecture timestamp: 01:12:56.

1.6.2 Reading and reversing a vertex

A single vertex represents a family of processes. Reversing an incoming line to an outgoing line changes a particle into its antiparticle, and reversing a W^- line gives a W^+. One can therefore read a vertex as emission, absorption, decay, annihilation, or the inverse reaction, provided charge, momentum, and all relevant quantum numbers are followed consistently.

Question from the audience. Can the arrows and lines in these weak vertices be flipped? [a]

Response. Yes. A line may be crossed from the final state to the initial state or vice versa, but a particle then becomes its antiparticle. The same local vertex consequently describes many differently oriented reactions.

[a]Lecture timestamp: 01:14:51.

Question from the audience. Does writing a W^- with strange and charm lines mean that the boson is literally a strange quark bound to an anti-charm quark? [a]

Response. No. The line bookkeeping says only that the W^- carries the quantum-number change between those columns. If the diagram is read as a decay, a sufficiently energetic W^- can produce a strange quark and an anti-charm quark; it is not composed of them.

[a]Lecture timestamp: 01:15:43.

Question from the audience. Can an electron and an antineutrino make a W, which then produces a quark pair? [a]

Response. Exactly. If enough center-of-mass energy is available, the initial leptons can form the charged-current intermediate state and that state can emerge through an allowed quark channel. The two vertices are the same ones used in decay, read in another orientation.

[a] Lecture timestamp: 01:17:22.

1.6.3 The missing third boson

An $SU(2)$ group has three generators, hence three gauge fields. So far we have displayed only W^+ and W^-. A third, electrically neutral field must be associated with diagonal combinations such as electron–positron and neutrino–antineutrino quantum numbers. It is deliberately left lurking in the background. Its relation to the physical Z boson and photon requires electroweak mixing, which comes later.

Question from the audience. Were the weak and strong forces suspected at different historical times? [a]

Response. Their experimental roots appeared together in early radioactivity. Alpha decay revealed nuclear and strong-interaction physics, beta decay revealed the weak interaction, and gamma decay emitted electromagnetic photons. Becquerel's era already contained all three clues, long before their modern field theories existed.

[a] Lecture timestamp: 01:24:11.

1.7 Worked Charged-Current Decays

1.7.1 Pion decay

The negatively charged pion has valence-quark content

$$\pi^- = d\bar{u}. \tag{1.39}$$

Those quantum numbers can annihilate into a W^-, which can then produce a charged lepton and the corresponding antineutrino:

$$\pi^- \longrightarrow W^{-*} \longrightarrow e^- + \bar{\nu}_e, \tag{1.40}$$

$$\pi^- \longrightarrow W^{-*} \longrightarrow \mu^- + \bar{\nu}_\mu. \tag{1.41}$$

The star reminds us that the intermediate W is virtual. The second channel is the dominant one.

Question from the audience. Is this really a Feynman diagram even though the time direction was drawn horizontally rather than vertically? [a]

Response. Yes. The orientation of time on the page is a convention. Once one direction is declared to be time and all arrows are interpreted consistently, the drawing carries the same interaction information.

[a] Lecture timestamp: 01:20:21.

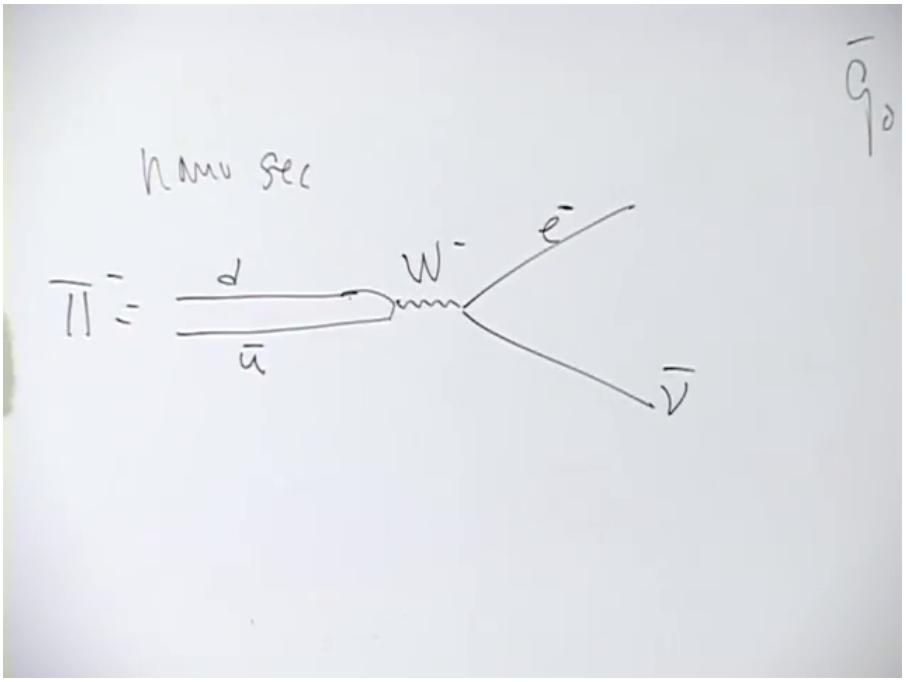

Figure 1.4: A $d\bar{u}$ pion annihilates into a virtual W^-, followed by a charged lepton and its matching antineutrino.

Lecture frame: 01:19:38.

Question from the audience. Could the diagram be read more like a reversible chemical reaction than as an irreversible one-way decay? [a]

Response. The inverse process is allowed. A pion produced in empty space decays because the outgoing particles separate and do not ordinarily recombine. But if a muon and antineutrino are fired together with the right energy and momentum, there is a nonzero amplitude to form the pion through the reverse interaction.

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[a] Lecture timestamp: 01:21:35.

1.7.2 Energy and a virtual intermediate state

For a pion at rest,

$$m_\pi \simeq 140 \text{ MeV}, \qquad m_\mu \simeq 106 \text{ MeV}, \qquad m_\nu \approx 0 \tag{1.42}$$

at the precision relevant here. The remaining energy becomes kinetic energy of the outgoing particles. Momentum conservation requires

$$\boldsymbol{p}_\mu + \boldsymbol{p}_{\bar{\nu}} = 0, \tag{1.43}$$

so the two particles leave back to back in the pion rest frame.

Question from the audience. How can energy be conserved when the intermediate W is far heavier than the pion? [a]

Response. Only the initial and final particles must be on their mass shell. Four-momentum is conserved at every vertex, but an internal virtual line need not obey $E^2 = \boldsymbol{p}^2 + m_W^2$. The larger its off-shell mismatch, the more strongly its propagator suppresses the process.[b]

[a] Lecture timestamp: 01:25:08.

[b] Editorial clarification: The lecture uses the traditional energy–time uncertainty language and says the intermediate state may violate energy briefly. The modern precise statement is the one in the response: energy-momentum is conserved, while a virtual internal line need not satisfy the real-particle mass-shell condition.

1.7.3 Muon decay

A muon can first change into its own neutrino by emitting a virtual W^-:

$$\mu^- \longrightarrow \nu_\mu + W^{-*}. \tag{1.44}$$

The virtual boson then produces an electron and electron antineutrino:

$$\mu^- \longrightarrow \nu_\mu + e^- + \bar{\nu}_e. \tag{1.45}$$

This is a purely leptonic process. The muon is about two hundred times heavier than the electron, so the final rest masses leave ample kinetic energy.

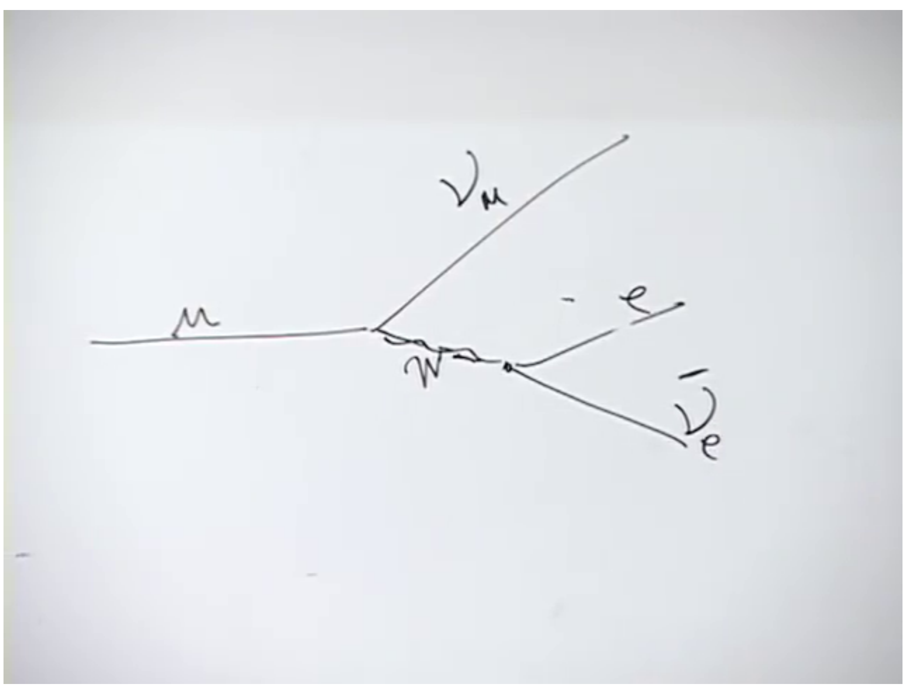

Figure 1.5: Muon decay through a virtual charged weak boson.

Lecture frame: 01:29:24.

Question from the audience. Is the neutral particle emitted with the electron an antineutrino? [a]

Response. Yes. The full final state contains one muon neutrino, one electron antineutrino, and the electron. Keeping the flavor labels prevents the two neutral particles from being confused.

[a]Lecture timestamp: 01:29:06.

The electron cannot reverse this decay in empty space because it is lighter than the muon. The tau is heavier and can decay through analogous weak channels. Stability here is partly an accident of the mass ordering, not a difference in the underlying vertex.

Question from the audience. Is a linear accelerator effectively increasing the electron's mass so that otherwise forbidden reactions can run in the opposite direction? [a]

Response. It does not change the electron's invariant rest mass, but it supplies kinetic energy. Sufficient center-of-mass energy can create heavier final particles, so the proposed physical idea is correct once mass is replaced by available collision energy.

[a]Lecture timestamp: 01:30:23.

1.7.4 Neutron beta decay at quark level

The neutron and proton valence compositions are

$$n = udd, \qquad p = uud. \tag{1.46}$$

One down quark in the neutron makes the charged-current transition

$$d \longrightarrow u + W^{-*}, \tag{1.47}$$

turning udd into uud. The virtual boson then decays:

$$W^{-*} \longrightarrow e^- + \bar{\nu}_e. \tag{1.48}$$

Combining the two vertices reproduces

$$n \longrightarrow p + e^- + \bar{\nu}_e. \tag{1.49}$$

The opposite quark transition $u \to d + W^+$ would leave a three-down-quark baryon, but that available state is too heavy for a free neutron to produce. The actual down-to-up route is the energetically allowed one.

Question from the audience. Could a neutron decay into a proton, a muon, and the corresponding antineutrino instead? [a]

Response. Not for a free neutron at rest: the neutron–proton mass difference is far smaller than the muon mass. With enough additional collision or excitation energy, the same weak vertex can participate in a channel that produces a muon. The prohibition is kinematic, not a missing interaction.

[a]Lecture timestamp: 01:32:38.

1.8 The Unresolved Suppression

The charged weak vertices explain which processes can occur, but not yet why they are so slow. One tempting answer would be an extraordinarily tiny weak coupling. That answer is wrong: the dimensionless weak gauge coupling is of roughly electromagnetic size. The suppression must come from something else.

The missing ingredient is already visible in the virtual propagator: the W is very massive. At energies far below m_W, charged-current amplitudes are suppressed roughly by powers of $1/m_W^2$. That connection will be developed after the Higgs mechanism and electroweak symmetry breaking explain how the gauge bosons acquire mass. [6]

Question from the audience. When we say that the theory has this symmetry, do we mean that its Lagrangian is invariant under the corresponding operations? [a]

Response. Exactly. The statement that the theory has the symmetry is shorthand for invariance of the Lagrangian, or equivalently the action, under those transformations. Writing the full Lagrangian is being postponed, not replaced by a different definition.

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[a] Lecture timestamp: 01:34:14.

The route is now complete: Maxwell theory supplies the prototype; QCD shows how non-Abelian gauge bosons carry and exchange charge; coupling constants explain the contrast between atomic and hadronic time scales; and weak $SU(2)$ organizes quarks and leptons into doublets connected by $W^{\pm}$. The remaining task is to understand why these gauge-mediated processes are weak in practice even though their underlying coupling is not exceptionally small.

[6] Editorial clarification: The lecture preserves this as the puzzle for the next meeting. The heavy-W explanation is included here as a forward connection, not as a claim that its derivation occurred in this lecture.